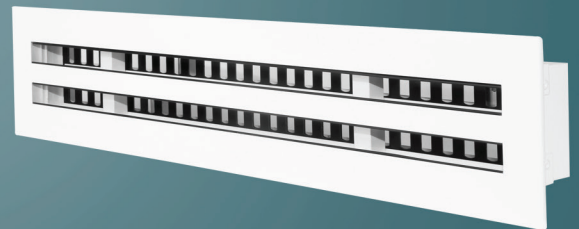
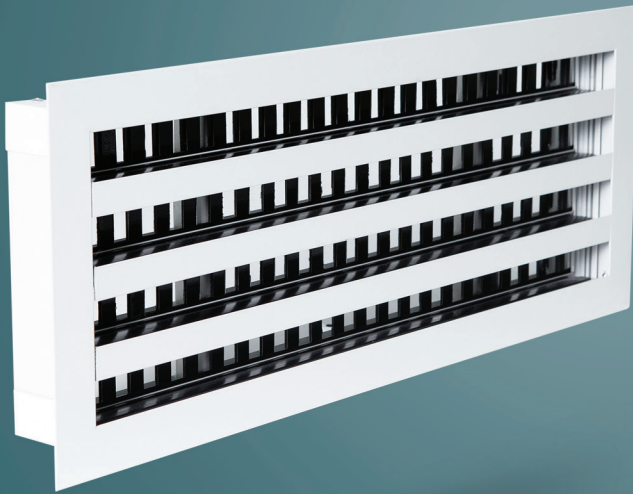
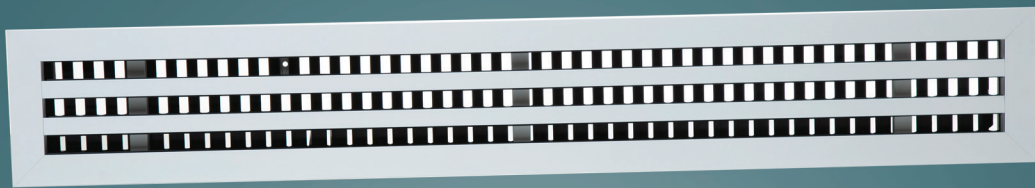




TECNALCO

CHAPTER 5



LINEAR SLOT DIFFUSER

CHAPTER-5

LINEAR SLOT DIFFUSERS

INDEX

- Supply,Return,Linear Slot Diffusers _____
- Construction Details and Dimensions _____
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SUPPLY, RETURN, DUMMY LINEAR SLOT DIFFUSERS

TECNALCO modulinear slot diffusers are specially designed to provide supply or return air in heating, cooling or ventilating systems. Constructed of extruded aluminium, these diffusers can be used in virtually any type of ceiling.

Available in 3 different slot widths, from 1 through 8 slots, modulinear diffusers permit complete control of flow rate without disturbing the selected air pattern.

Available in 2 defferent types:

- 1) TECNALCO standard type 'TS'
- 2) TECNALCO type TS-(EF)
- 3) TECNALCO 'ER' type 'T'ER'

Features:

- All types of models are designed to give full flexibility in volume and air pattern control.
- Single or multiple slot arrangement will provide full 180° adjustable air diffusing patterns.
- Air diffusion pattern can be made horizontal along the ceiling, vertically downwards or intermediate.
- Deflecting vanes and integral volume damper can be adjusted from the face of the diffuser. By adjusting the integral damper flow volume can be adjusted. The flow control does not affect the air pattern.
- The integral volume control damper is designed in a unique way that it can be used as an equalizing grid.
- Positive alingment of adjacent sections using alignment strips, provided with each diffuser.
- The diffuser is constructed from high quality aluminium extrusions.

Material construction:

- Slot diffuser is made of extruded aluminium in natural anodised finish.
- Face section is available in powder coated finish also.
- Air pattern controlling deflectors (Hit and miss damper) are in black matt finish.
- Integral volume damper is in mill finish, available on request in black matt finish.
- Plenum is made of galvanised sheet steel and internal insulation (acoustic liner) is of mineral fibre are optional.
- Balancing damper, made of galvanised sheet steel.

Types available:

Type	Supply	Return	Dummy
T'S'	SLD	RLD	DLD
TS-EF	SLD	RLD	DLD
T'ER'	SLD	RLD	DLD

Standard sizes:

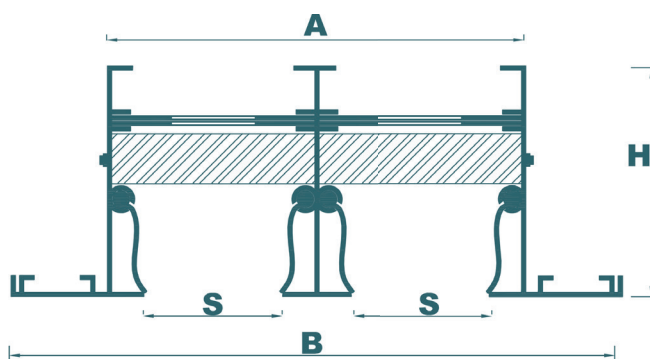
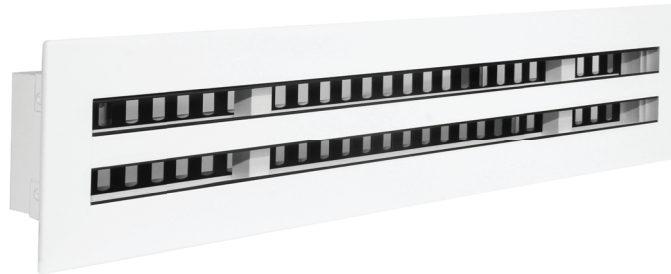
No. of Slots	In Number of Pieces		
	1 pc.	2 pcs.	3 pcs.
1, 2 & 3 slots	≤ 4 met.	> 4 met.	> 6 met.
4 slots	≤ 4 met.	> 4 met.	> 6 met.
5 slots	≤ 3.5 met.	> 3.5 met.	> 5 met.
6 slots	≤ 3.5 met.	> 3.5 met.	> 5 met.
7 slots	≤ 3 met.	> 3 met.	> 4.5 met.
8 slots	≤ 2.5 met.	> 2.5 met.	> 4.5 met.

NOTE:

Each piece incorporates brackets and strips for alignment and suspension purposes.

SUPPLY, RETURN, DUMMY LINEAR SLOT DIFFUSERS

TECNALCO Model T-S Type



TECNALCO Standard Models - 'TS':

TECNALCO 'ER' Models:

Model	'TL'	'TM'	'TH'
T'S' Standard	1-8 Slots	1-8 Slots	1-8 Slots
T'S'-EF	1-8 Slots	1-8 Slots	1-8 Slots
T'ER'	1-8 Slots	1-8 Slots	1-8 Slots

- 1) Model 'TL' : 16 mm slot width
- 2) Model 'TM' : 20 mm slot width
- 3) Model 'TH' : 25 mm slot width

More than 8 slots → available on request.

Available from 1 through 8 slots with 3 different slot widths for low, medium and high capacity requirements.

SLD 'TS'

TECNALCO model SLD 'TS' is a supply linear slot diffuser with integral volume control damper also called as (Hit and miss damper) and set of deflection blades that are fully adjustable from the face of the diffuser.

RLD 'TS'

TECNALCO model RLD 'TS' is a return or extract linear slot diffuser with air pattern controlling deflector blades. These are fully adjustable from the face of the diffuser.

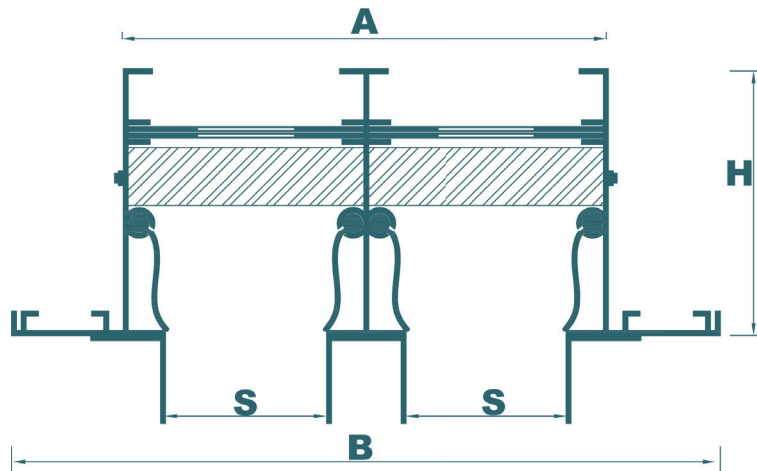
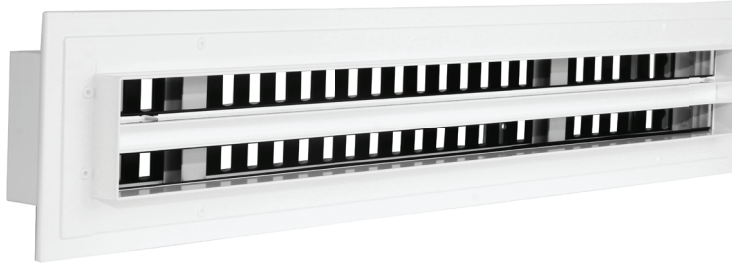
* single hit and miss damper provided on option.

DLD 'TS'

TECNALCO model DLD 'TS' is a dummy or non active linear slot diffuser with deflector blades and a blanking plate to replace hit and miss dampers.

SUPPLY, RETURN, DUMMY LINEAR SLOT DIFFUSERS

TECNALCO Model TS-EF (Extra Flange) Type



- 1) Model 'TL' : 16 mm slot width
- 2) Model 'TM' : 20 mm slot width
- 3) Model 'TH' : 25 mm slot width

More than 8 slots → available on request.

Available from 1 through 8 slots with 3 different slot widths for low, medium and high capacity requirements.

SLD 'TS'-EF

TECNALCO model SLD 'TS'-EF is a supply linear slot diffuser with extra flange on the face and integral volume control damper and hit and miss air straightening deflectors. Hit and miss air straighteners and integral damper are fully adjustable from the face of the diffuser.

RLD 'TS'-EF

TECNALCO model RLD 'TS'-EF is a return or extract linear slot diffuser with air pattern controlling deflector blades. These are fully adjustable from the face of the diffuser.

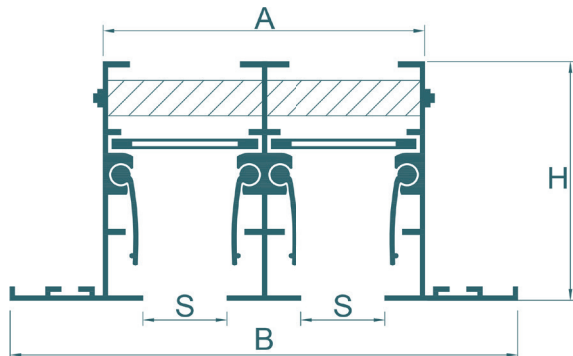
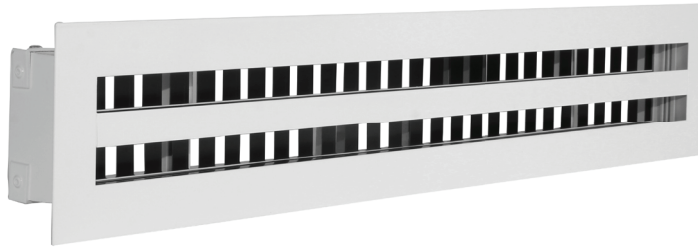
* single hit and miss damper provided on option.

DLD 'TS'-EF

TECNALCO model DLD 'TS'-EF is a dummy or non active linear slot diffuser with deflector blades a blanking plate to replace hit and miss dampers.

SUPPLY, RETURN, DUMMY LINEAR SLOT DIFFUSERS

TECNALCO Model T-ER Type



For T'S' and T'ER' type models:

- N = Nominal size = Duct size = L x W
- N = L x No. of Slots
- W = Width of the plenum, indicated in No. of Slots
- L = Length of the Slot Diffuser
- A = Neck size = Width of the slot diffuser neck
- B = Overall width from the face side of the slot diffuser (flange to flange width)
- S = Slot width = 16 mm / 20 mm / 25 mm of models TL / TM / TH respectively

SLD 'TER'

TECNALCO model SLD 'TER' is a supply linear slot diffuser with integral volume control damper and hit and miss air straightening deflectors. Hit and miss air straighteners and integral damper are fully adjustable from the face of the diffuser.

RLD 'TER'

TECNALCO model RLD 'TER' is a return or extract linear slot diffuser with air pattern controlling deflectors (Hit and miss damper). Hit and miss damper is fully adjustable from the face of the diffuser.

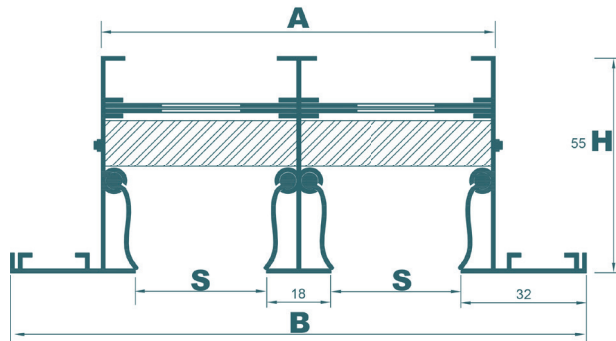
DLD 'TER'

TECNALCO model DLD 'TER' is a dummy or non active linear slot diffuser with hit and miss damper and a blanking plate to replace integral damper.

Coding:

1. Slot diffuser
2. Slot width
3. Integral damper
4. Hit and miss damper
5. Diffuser flange

CONSTRUCTION DETAILS AND DIMENSIONS T-S TYPE



Slot diffuser dimensions:

TECNALCO Standard 'TS' type models:

No. of Slots	Model 'TL' Slot 16 mm		Model 'TM' Slot 20 mm		Model 'TH' Slot 25 mm	
	A	B	A	B	A	B
1	37	80	41	84	45	89
2	70	114	78	122	88	132
3	104	148	116	160	131	175
4	138	182	154	198	174	218
5	172	216	192	236	217	261
6	206	250	230	274	260	304
7	240	284	268	312	304	347
8	274	318	306	350	347	390

N = Nominal size = L x W

L = Length of linear slot diffuser

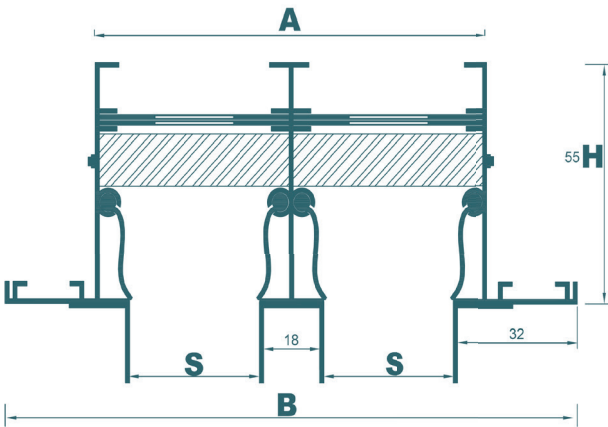
W = Width of linear slot diffuser

indicated in No. of Slots

• False ceiling opening size = (B - 30) mm.

• Error correction = $\pm 1\%$ due to manual work.

CONSTRUCTION DETAILS AND DIMENSIONS T-S-EF TYPE



Slot diffuser dimensions:

TECNALCO Standard TS-EF type models:

No. of Slots	Model 'TL' Slot 16 mm		Model 'TM' Slot 20 mm		Model 'TH' Slot 25 mm	
	A	B	A	B	A	B
1	37	80	41	84	45	89
2	70	114	78	122	88	132
3	104	148	116	160	131	175
4	138	182	154	198	174	218
5	172	216	192	236	217	261
6	206	250	230	274	260	304
7	240	284	268	312	304	347
8	274	318	306	350	347	390

N = Nominal size = L x W

L = Length of linear slot diffuser

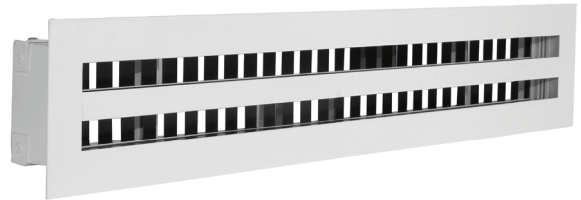
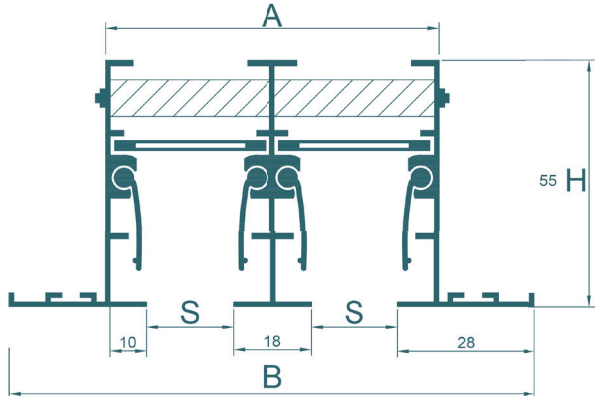
W = Width of linear slot diffuser

indicated in No. of Slots

• False ceiling opening size = (B - 30) mm.

• Error correction = $\pm 1\%$ due to manual work.

CONSTRUCTION DETAILS AND DIMENSIONS T-ER TYPE



Slot diffuser dimensions:

TECNALCO Standard 'ER' type models:

No. of Slots	Model 'TL' Slot 16 mm		Model 'TM' Slot 20 mm		Model 'TH' Slot 25 mm	
	A	B	A	B	A	B
1	36	72	40	76	45	81
2	70	106	78	114	88	126
3	104	140	116	152	131	167
4	138	174	154	190	174	210
5	172	208	192	228	217	253
6	206	242	230	266	260	296
7	240	276	268	304	303	339
8	274	310	306	342	346	382

N = Nominal size = L x W

L = Length of linear slot diffuser

W = Width of linear slot diffuser
indicated in No. of Slots

• False ceiling opening size = (B - 26) mm.

• Error correction = $\pm 1\%$ due to manual work.

CONSTRUCTION DETAILS AND DIMENSIONS

Type T'S' & TS-EF

Type T'ER'



● A → with End Cap



● B → with End Flange



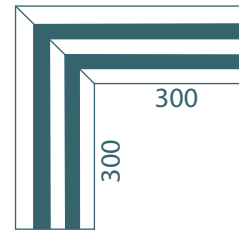
● C → with open ends



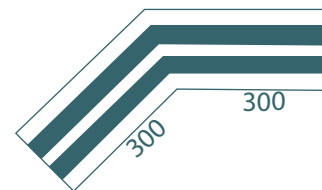
● D → with one side End Cap/Flange (DA/DB)



● E with both side End Caps/Flanges (EA/EB)



● 90° corner piece



● 135° corner piece



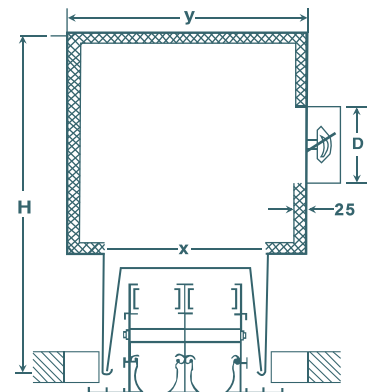
SLD-T'S'-T'M'-DA

SLD-T'S'-T'M'-C

SLD-T'S'-T'M'-DA

For positive alignment of adjacent sections alignment strips provided with each diffuser.

CONSTRUCTION DETAILS AND DIMENSIONS



Construction Details and Dimensions

Plenum box details:

- TECNALCO acoustically lined plenum box is used for evenly distributing cool air over the surface of the linear diffuser with minimum noise turbulence, thereby maintaining correct performance characteristics. It saves fabrication and installation labour.
- Plenums are manufactured to suit the full range of linear diffusers from 1 to 8 slots.
- All joints are welded and sealed for air tight functioning which adds to maintain rigidity.
- Internal baffles are perforated sheet or splitters to suit the design, or a combination of both in special cases.
- Maximum one piece length is 1500 mm. Above this length plenum are supplied in multiple sections of ever length each provided with an intel spigot.

Material:

Plenum box is manufactured from galvanised sheet steel of 0.7 mm thickness and internally lined insulation is of 25 mm thickness mineral fibre.

Plenum dimensions with insulation:

No. of Slots	DØ	H	x	y
1	150	300	*A + 10	x + 100
2	200	300	*A + 10	x + 100
3	200	300	*A + 10	x + 100
4	250	300	*A + 10	x + 100
5	250	300	*A + 10	x + 100
6	315	365	*A + 10	x + 100
7	315	365	*A + 10	x + 100
8	315	365	*A + 10	x + 100

NOTE:

- * A = slot diffuser neck size
Thickness of the insulation = 25 mm

Plenum dimensions without insulation:

No. of Slots	DØ	H	x	y
1	150	275	*A + 10	x + 50
2	200	275	*A + 10	x + 50
3	200	275	*A + 10	x + 50
4	250	275	*A + 10	x + 50
5	250	275	*A + 10	x + 50
6	315	340	*A + 10	x + 50
7	315	340	*A + 10	x + 50
8	315	340	*A + 10	x + 50

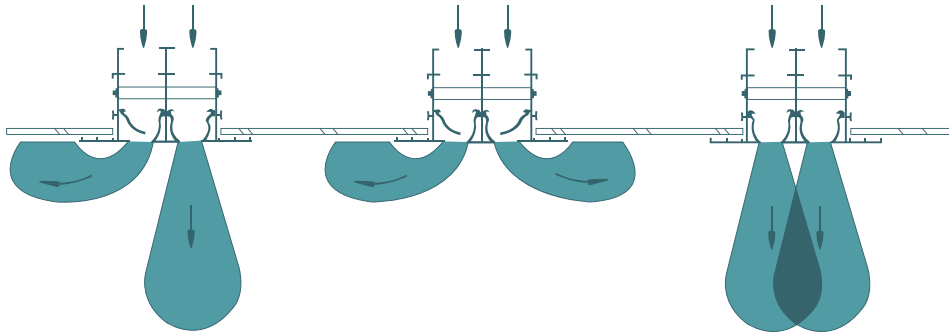
NOTE:

- * A = slot diffuser neck size

ENGINEERING AND PERFORMANCE DATA

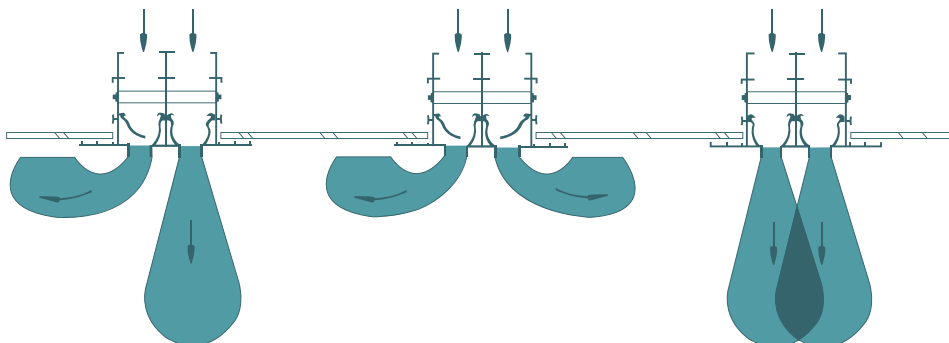
Possible combinations of discharging of air by adjusting the Hit and miss damper.

With Type SLD T'S' :



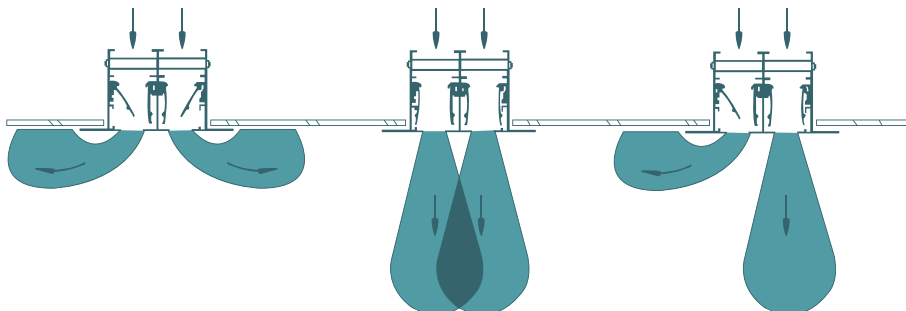
- 50% horizontal along the ceiling 50% vertically downwards
- 100% horizontal along the ceiling
- 100% vertically downwards

With Type SLD TS-EF :



- 50% horizontal along the ceiling 50% vertically downwards
- 100% horizontal along the ceiling
- 100% vertically downwards

With Type SLD T'ER' :



- 100% horizontal along the ceiling 50% vertically downwards
- 100% vertically downwards
- 50% horizontal along the ceiling 50% vertically downwards

ENGINEERING AND PERFORMANCE DATA

Air Induction

In ceiling application, the air induction ratio for slot diffusers without any blade setting is low, compared to air induction with ceiling diffusers.

Air induction increases very rapidly with the adjustment of the deflection blades.

For such reason, no deflection should be applied with false ceiling height of more than 4.00 meter.

For low ceiling height the deflection becomes necessary and throw must be adapted accordingly.

With slot diffusers, flow pattern can be adjusted to any local situation.

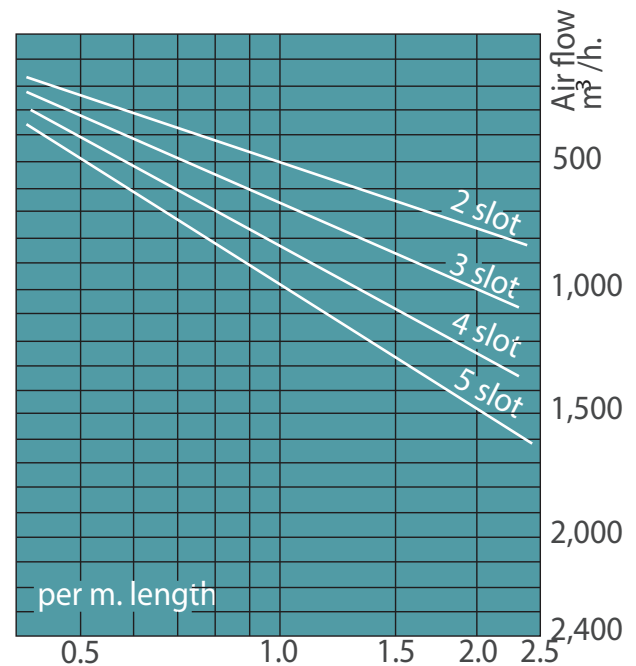
Influence on air induction with and without blade setting is shown on opposite graphs and photos are made under following conditions:

- three-slot diffuser TECNALCO model SLD 'TS' 'TM'
- standard model with 20 mm slot width
- no deflection setting
- deflection blades adjusted to 45° angle setting.
- no restriction on air flow
- diffuser length 1000 mm

This model gives an air flow pattern which is adjustable from full left to full right by simply adjusting the deflection blades (Hit and miss damper). This can be easily done before or after installation.

This high capacity unit has been designed to serve the performance oriented market of today.

Air flow capacity can be controlled by the integral volume damper or intel spigot damper in the plenum box.



Induced air ratio to m³/h supply air.

Technical data

Induced Air Flow

Symbols

v_0	Outlet air velocity	m/s
v_t	Terminal air velocity	m/s
V	Supply air flow rate	m³/h or m³/s
V_i	Induced air flow rate	m³/h or m³/s
V_t	Total air volume at end of throw	m³/h or m³/s
	$V_L = V + V_i$	
	Induction ratio	
t_0	Supply air temperature	°C
t_r	Room temperature	°C
t_x	Air temperature in the jet axis at a distance 'x'	°C
Δt_0	Supply air temperature differential	K
Δt_x	Temperature differential at a distance 'x'	K
$\lambda(\Delta t)$	Δt ratio	

Formulae

$$i = \frac{V_L}{V_i}$$

$$\Delta t_x = 0.8 \cdot \Delta t_0 \left(\frac{V_x}{V_0} \right)$$

$$\lambda(\Delta t) = \frac{\Delta t_x}{\Delta t_0}$$

$$\Delta t_x = t_r + \Delta t_0 \cdot \lambda(\Delta t)$$

$$\Delta t_0 = t_0 - t_r$$

$$\Delta t_x = t_x - t_r$$

ENGINEERING AND PERFORMANCE DATA

Sound level

The sound level of a linear slot diffuser is in direct ratio to the velocity of the air passing through it. due to the reduced sound level of SLD diffusers, air passing through a properly selected outlet will not add any appreciable noise to the sound level of the general air conditioning system. When hit and miss damper or integral damper is partially closed for balancing purposes, noise level can increase in function of the importance of the air distributing network. For distributing network extending more than 50 meter, basic balancing should be made by volume control dampers in the ducting. adjustment of deflection blades does not add any complementary sound level. Final sound attenuation in the distributing system can be obtained inside sound insulation in the plenum boxes.

In order to obtain reasonable sound level, velocity in the outlet should be selected as follows:

Ceiling height

3 m - max 3 m/sec.

4 m - max 4 m/sec.

4.5 m. - max 5 m/sec.

Effect of damper adjustment on noise generation

Symbols

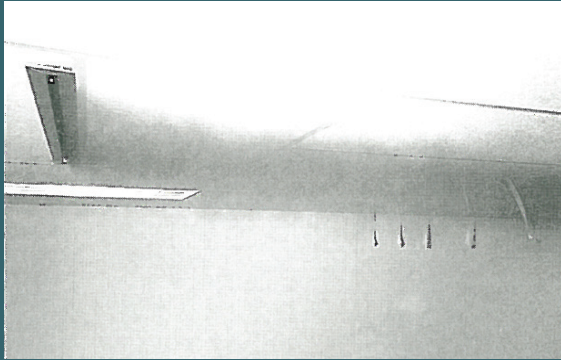
$\Delta P1$	Total pressure drop through register, with damper fully open (read from selection diagram)	Pa
$\Delta P2$	Total pressure drop through register, with damper partially open	Pa
$\frac{\Delta P1}{\Delta P2}$	Ratio of pressure drops	
NR1	Noise level with damper fully open (c.f. selection diagram)	
NR2	Noise level with damper partially open	
ΔNR	Increase in noise level	

Formula

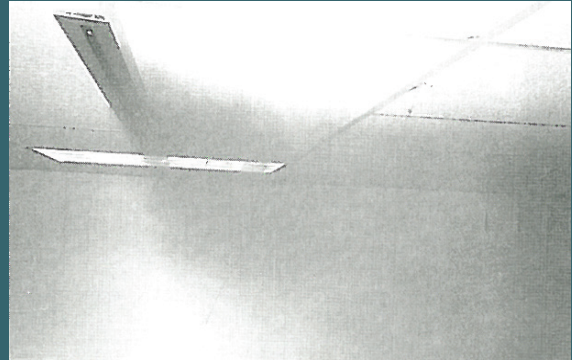
$$NR2 = NR1 + \Delta NR$$

ENGINEERING AND PERFORMANCE DATA

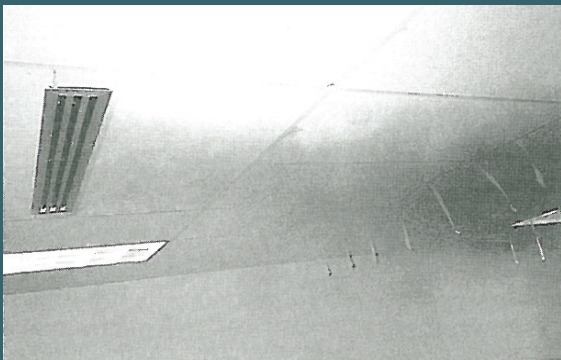
Discharging of air pattern can be observed from the smoke test results.



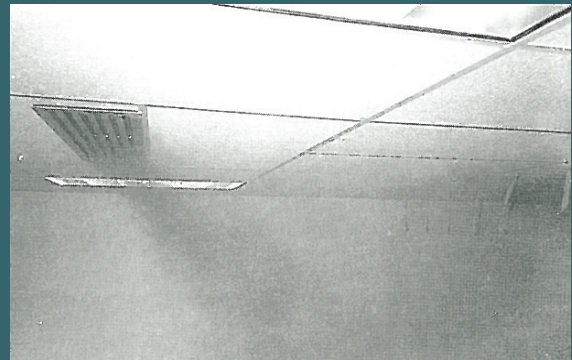
Model: SLD-T'S'-T'M' 1 slot x 1000 mm
Discharging of air horizontal along the ceiling



Model: SLD-T'S'-T'M' 1 slot x 1000 mm
Discharging of air 45° downwards



Model: SLD-T'S'-T'M' 3 slot x 1000 mm
Discharging of air horizontal along the ceiling



Model: SLD-T'S'-T'M' 6 slot x 1000 mm
Discharging of air 45° downwards

Smoke tests conducted by ETL Testing Laboratories Inc. An ADC certified lab.

ENGINEERING AND PERFORMANCE DATA

Test Results from ETL

The tests were conducted, by the ETL Testing Laboratories on the selected sizes of the Linear slot Diffusers. The Linear Slot Diffusers had nominal 20 mm wide slots. Each slot was equipped with 2 deflector vanes (Hit and Miss Damper) which could be adjusted to provide air flow in any direction from horizontal to vertical. The intel to each slot contained a sloted slide damper (Integral volume damper). The dampers were set to the full open position for all tests. The test results include Total pressure, Area factor, Throw pattern and Soundpower level.

Test method : The diffusers were tested in accordance with Air Diffusion Council Test Code for Grilles, Registers and Diffusers ADC 1062: GRD-84. The diffusers were installed in the ADC facility and supplied with measured volumes of air. The static Pressure was measured 1-1/2 duct diameters upstream of the diffuser intel.

Throw T : Air volumes are given in Lt/sec.

NC values : The throw values given in the tables are related to the ceiling diffuser placed nearer to the ceiling. Three throw values are given for each q_v to the corresponding terminal velocities V_t .

NC values : Minimum Throw $\rightarrow 0.75$ m/sec. V_t

Middle Throw $\rightarrow 0.50$ m/sec. V_t

Maximum Throw $\rightarrow 0.25$ m/sec. V_t

NC values : Noise data is given in terms of NC levels. These have been derived from sound power level data (dB re 10^{-12} watts) by subtracting 10 dB (Room effect) from the sound power levels.

Outlet size : Outlet sizes are mentioned on the left side of each row in terms of number of slots x 1 m length.

Linear Slot Diffusers Correction factors for other lengths:

L (m)	1	2	3 - 5	6 - 10
Throw (m)	x 1	x 1.05	x1.1	x 1.15
N.C.	+ 0	+ 1	+ 4	+ 8

Area Factor A_k

Manufacturer: tecnalco								Outlet/inlet type & size: 305 mm diameter				
Product: linear slot diffuser - 6 slot								Neck area: 729 cm ²				
Flow meter equipment: 305 mm metering station								Anemometer: alnor velometer type 6000p				
152 mm sharp edged orifice								Type & serial number: 6077al				
Run No.	READINGS							Calculations				
	Flow Meter Δp pa	Discharge velocity in m/sec.					Static Pressure Pa	Flow l/s	Neck Vel. M/s	Neck V.P. P/a	Total Press. P/a	Area factor (A_k)
		1	2	3	4	Ave.						
1	24.89	1.88	1.73	1.73	1.73	1.77	3.24	75	1.03	0.75	3.98	0.042
2	44.80	2.44	2.34	2.34	2.29	2.35	5.72	100	1.37	1.24	6.97	0.043
3	69.69	3.10	2.90	2.90	2.90	2.95	8.96	125	1.72	1.74	10.70	0.042
Average												0.042

ENGINEERING AND PERFORMANCE DATA

Throw Test Data

Product: Linear Slot Diffuser - 6 Slot

Run No. 1 Air Volume L/s 75

Distance from Outlet (m)

cm	0	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0	5.5	6.0
2.5	1.85	1.89	0.91	0.86	0.71	0.66	0.53	0.33	-				
7.5	1.07	1.04	0.84	0.74	0.58	0.53	0.46	0.30	-				
15	-	1.04	0.89	0.74	0.56	0.48	0.43	0.28	-				
22	-	0.79	0.91	0.74	0.56	0.51	0.43	0.28	-				
30		-	0.53	0.48	0.41	0.36	0.30	-					
45		-	0.46	0.51	0.46	0.48	0.38	-					
60			-	-	0.25	-	-						
90					-								
120													

Velocity in m/s.

Run No. 2 Air Volume L/s 100

Distance from Outlet (m)

cm	0	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0	5.5	6.0
2.5	2.21	1.52	1.32	1.14	0.94	0.81	0.71	0.61	0.46	0.36	0.28	-	
7.5	1.68	1.42	1.22	1.02	0.76	0.66	0.61	0.53	0.41	0.33	0.25	-	
15	-	1.37	1.27	1.04	0.74	0.58	0.58	0.53	0.36	0.30	-		
22	-	1.07	1.24	0.99	0.74	0.56	0.58	0.53	0.38	0.30	-		
30	-	0.25	0.74	0.61	0.53	0.43	0.43	0.38	0.28	0.25	-		
45		-	0.51	0.56	0.69	0.53	0.46	0.46	0.33	0.25	-		
60			-	0.28	0.33	0.38	0.30	0.30	0.28	0.28	-		
90				-	-	-	-	0.28	-	-			
120								-					

Velocity in m/s.

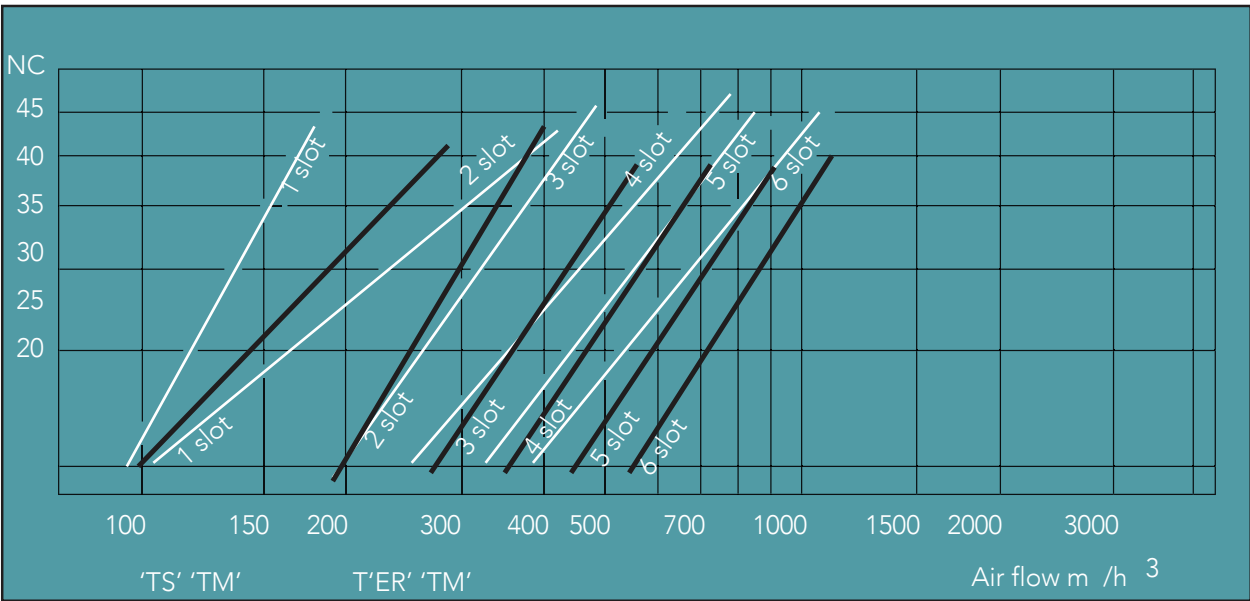
ENGINEERING AND PERFORMANCE DATA

Sound Test Data

Octave Band Center Frequency Hertz	Linear Slot Diffuser - 6 Slot			
	Discharge Sound Power Level Lw dB re 10 ⁻¹² Watt			
	1	2	3	4
125	49.5	55.0	60.5	64.0
250	41.5	47.5	53.5	58.0
500	37.0	45.0	52.0	56.5
1000	29.0	38.5	47.0	52.5
2000	23.5	33.0	44.0	50.5
4000	21.5*	23.5*	35.0	43.5
8000	27.5*	27.5*	27.5*	29.0*
Total Air Volume L/s	150	200	250	300
Intel Total Pressure, Pa	15	26	41	58
**Noise Criteria	21	30	37	42

- * Sound power level data marked with an asterisk has reached ambient levels in the test room. Actual levels are less than or equal to the levels indicated.
- ** Noise Criteria ratings were determined by subtracting 10 dB (room effect) from the Sound Power Levels.

Noise level for Linear Diffusers Model T'S' 'TM',TS-EF/T 'ER' 'TM'



ENGINEERING AND PERFORMANCE DATA

Selection tables for Linear Slot Diffusers
Types T'S, TS-EF and TER' of model 'TL' - slot width = 16 mm

No. of Slots	Area Factor A_k	Liters / Second / Meter												
		30	40	50	75	100	125	150	200	250	300			
1	0.0056	Pt (Pa) Throw (m) NC 20 3.6-5.4-7.8 21	36 5.4-6-9.6 26	56 6-7.2-11.4 33	127 7.2-9-14.4 46									
2	0.0112	Pt (Pa) Throw (m) NC 10 3.6-4.8-8.4 <20	16 4.8-6.6-10.2 21	35 6.6-8.4-13.2 33	62 7.8-10.2-15.6 42									
3	0.0168	Pt (Pa) Throw (m) NC 7 3.6-6-8.4 <20		16 5.4-8.4-12 22	29 7.2-9.6-15 30	44 9-11.4-16.8 37	62 10.2-12.6-19.2 43							
4	0.0224	Pt (Pa) Throw (m) NC 10 4.2-7.2-12 <20			15 6.6-9-14.4 23	27 7.8-11.4-16.2 31	40 9.6-12-18 38	70 12-14.4-20.4 44						
5	0.028	Pt (Pa) Throw (m) NC 11 4.8-7.2-13.8 <20			19 6.6-10.8-16.2 23	26 8.4-12-17.4 32	46 11.4-14.4-20.4 39	72 13.2-16.8-25.2 44						
6	0.0336	Pt (Pa) Throw (m) NC 12 5.4-9.6-16.2 <20			19 6.6-10.8-16.2 23	26 8.4-12-17.4 32	46 11.4-14.4-20.4 39	72 13.2-16.8-25.2 44						
7	0.0392	Pt (Pa) Throw (m) NC 14 6-10.8-17.4 21			14 6-10.8-17.4 21	25 10.2-13.8-20.4 29	39 12-15.6-23.4 37	56 13.8-16.8-25.8 43						
8	0.0448	Pt (Pa) Throw (m) NC 11 4.8-10.8-16.8 <20			11 4.8-7.2-13.8 <20	19 6.6-10.8-16.2 23	26 8.4-12-17.4 32	46 11.4-14.4-20.4 39						

ENGINEERING AND PERFORMANCE DATA

Selection tables for Linear Slot Diffusers
Types TS', TS-EF and TER' of model 'TL' - slot width = 20 mm

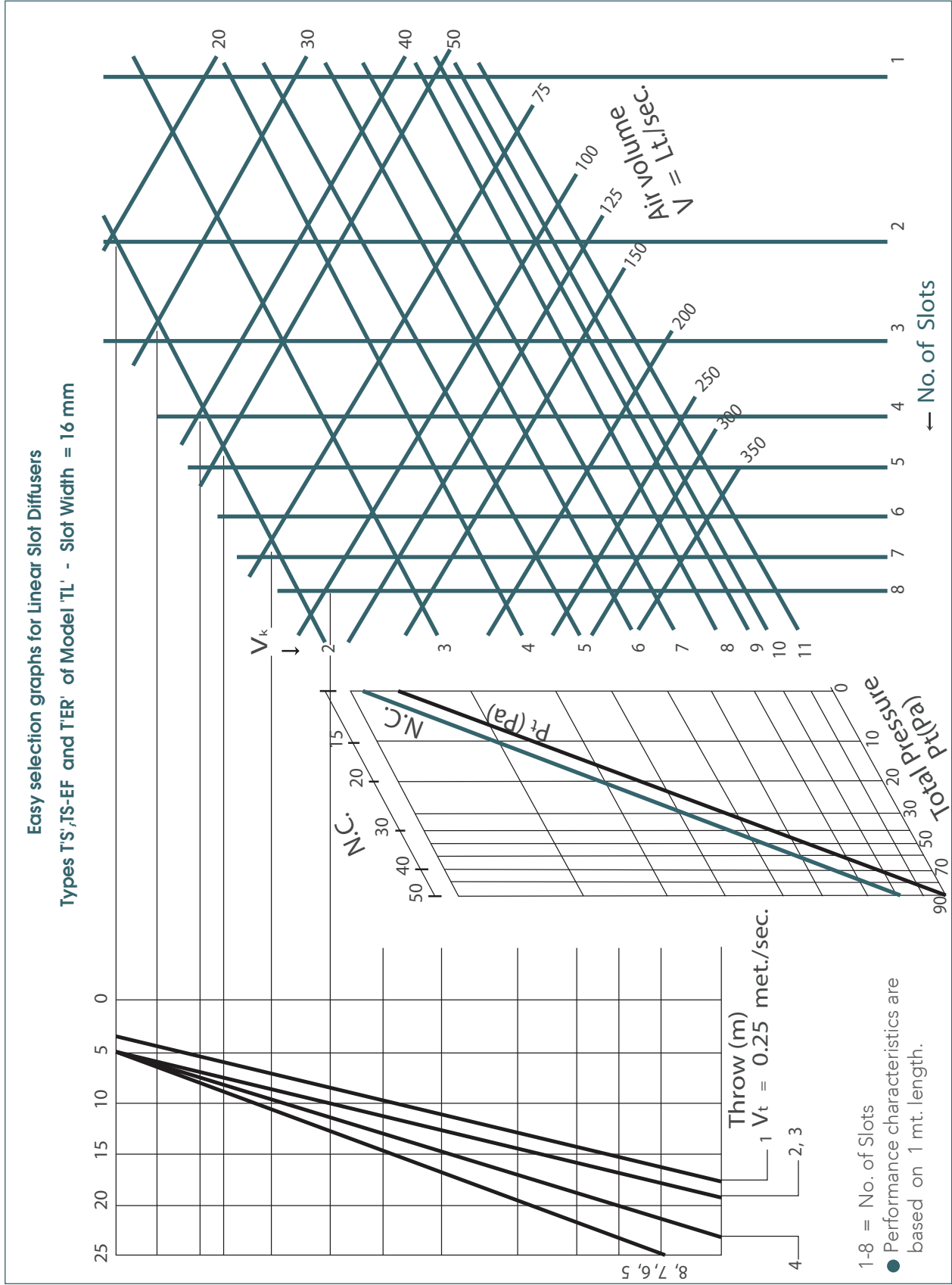
No. of Slots	Area Factor Ak	Liters / Second / Meter									
		30	40	50	75	100	125	150	200	250	300
1	0.007	Pt (Pa) Throw (m) NC 16 3.0-4.5-6.5 17	29 4.5-5.8-8.0 22	45 5.0-6.0-9.5 29	102 6.0-7.5-12.0 42						
2	0.014	Pt (Pa) Throw (m) NC 8 3.0-4.0-7.0 15	13 4.0-5.5-8.5 17	28 5.5-7.0-11.0 29	50 6.5-8.5-13.0 38						
3	0.021	Pt (Pa) Throw (m) NC 6 3.0-5.0-7.0 15	13 4.5-7.0-10.0 18	23 6.0-8.0-12.5 26	35 7.5-9.5-14.0 33	50 8.5-10.5-16.0 39					
4	0.029	Pt (Pa) Throw (m) NC 8 3.5-6.0-10.0 15	14 5.5-7.5-12.0 19	22 6.5-9.5-13.5 27	32 8.0-10.0-15.0 39	56 10.2-12.0-17.0 40					
5	0.036	Pt (Pa) Throw (m) NC 9 4.0-6.0-12.0 15	15 5.5-9.0-13.5 19	21 7.0-10.0-14.5 28	37 9.5-12.0-17.0 35	58 11.0-14.0-21.0 40					
6	0.042	Pt (Pa) Throw (m) NC 10 4.5-8.0-13.5 15	15 6.5-9.5-14.5 21	26 9.0-12.0-17.0 30	41 10.5-13.5-20.0 37	58 12.0-14.5-22.0 42					
7	0.050	Pt (Pa) Throw (m) NC 11 5.0-9.0-14.5 17	17 5.0-9.0-14.5 17	20 8.5-11.5-17.0 25	31 10.0-13.0-19.5 33	45 11.5-14.0-21.5 39					
8	0.057	Pt (Pa) Throw (m) NC 9 4.0-9.0-14.0 15	16 8.0-11.0-16.0 21	24 10.0-13.0-19.0 29	35 11.0-14.0-21.0 36						

ENGINEERING AND PERFORMANCE DATA

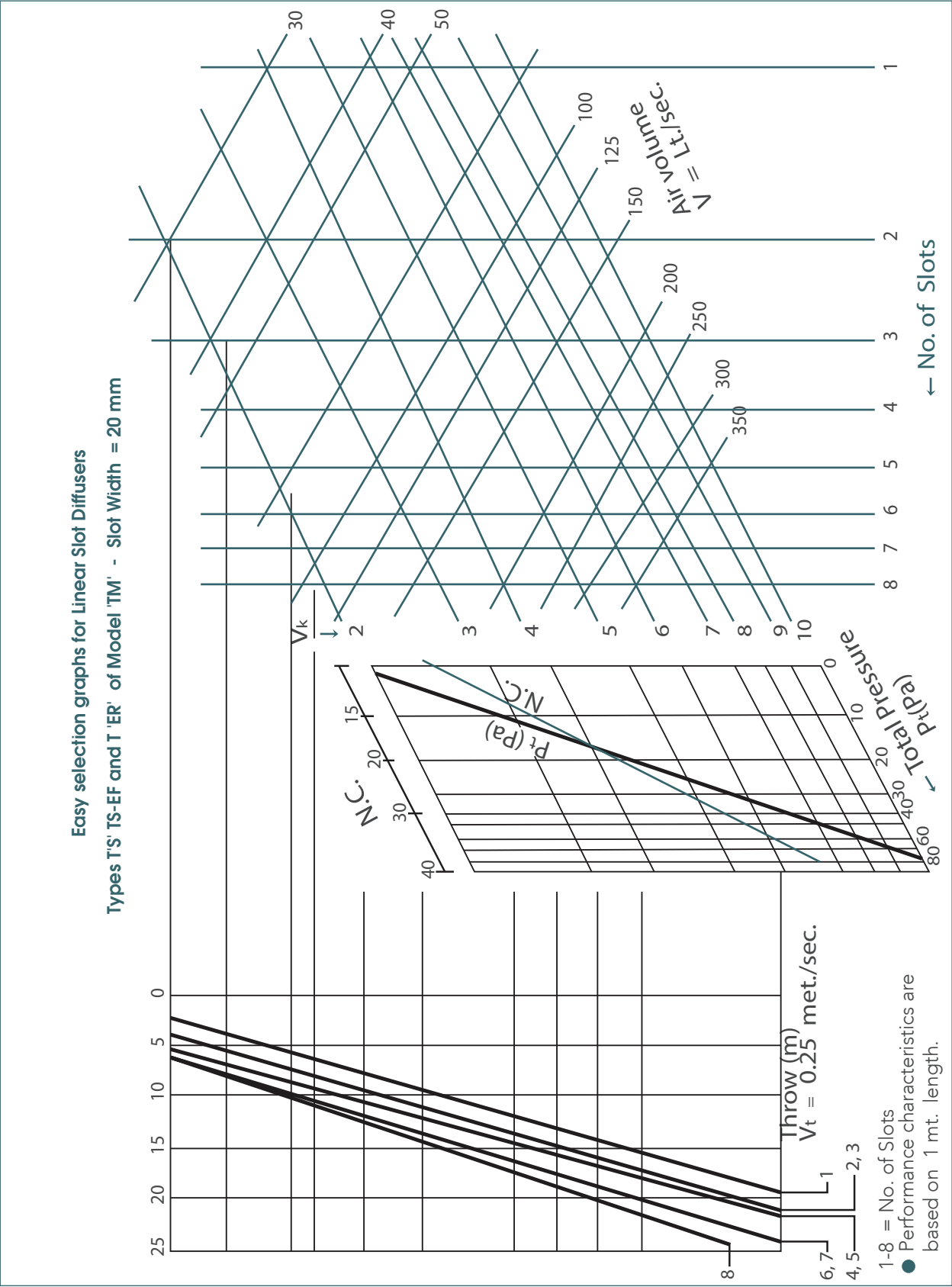
Selection tables for Linear Slot Diffusers
Types TS, TS-EF and TER of model 'TH' - slot width = 25 mm

No. of Slots	Area Factor A_k	Liters / Second / Meter												
		30	40	50	75	100	125	150	200	250	300			
1	0.009	Pt (Pa) Throw (m) NC	13 2.4-3.6-5.2 15	23 3.6-4-6.4 19	36 4-4.8-7.6 26	82 4-6-9.6 39								
2	0.018	Pt (Pa) Throw (m) NC		6 2.4-3.2-5.6 15	10 3.2-4.4-6.8 15	22 4.4-5.6-8.8 26	40 5.2-6.8-10.4 35							
3	0.027	Pt (Pa) Throw (m) NC			5 2.4-4-5.6 15	10 3.6-5.6-8 15	18 4.8-6.4-10 23	28 6-7.6-11.2 30	40 6.8-8.4-12.8 36					
4	0.036	Pt (Pa) Throw (m) NC				6 2.8-4.8-8 15	11 4.4-6-9.6 16	18 5.2-7.6-10.8 24	26 6.4-8-12 31	45 8-9.6-13.6 37				
5	0.045	Pt (Pa) Throw (m) NC					7 3.2-4.8-9.6 15	12 4.4-7.2-10.8 16	17 5.6-8-11.6 25	30 7.6-9.6-13.6 32	47 8.8-11.2-16.8 37			
6	0.054	Pt (Pa) Throw (m) NC						8 3.6-6.4-10.8 15	12 5.2-7.6-11.6 18	21 7.2-9.6-13.6 27	33 8.4-14.3-16 34	46 9.6-11.6-17.6 39		
7	0.063	Pt (Pa) Throw (m) NC							9 4-7.2-11.6 15	16 6.8-9.2-13.6 22	25 8-10.4-15.6 30	36 9.2-11.2-17.2 36		
8	0.072	Pt (Pa) Throw (m) NC							7 3.2-7.2-11.2 15	13 6.4-8.8-12.8 18	19 8-10.4-15.2 26	28 8.8-11.2-16.8 33		

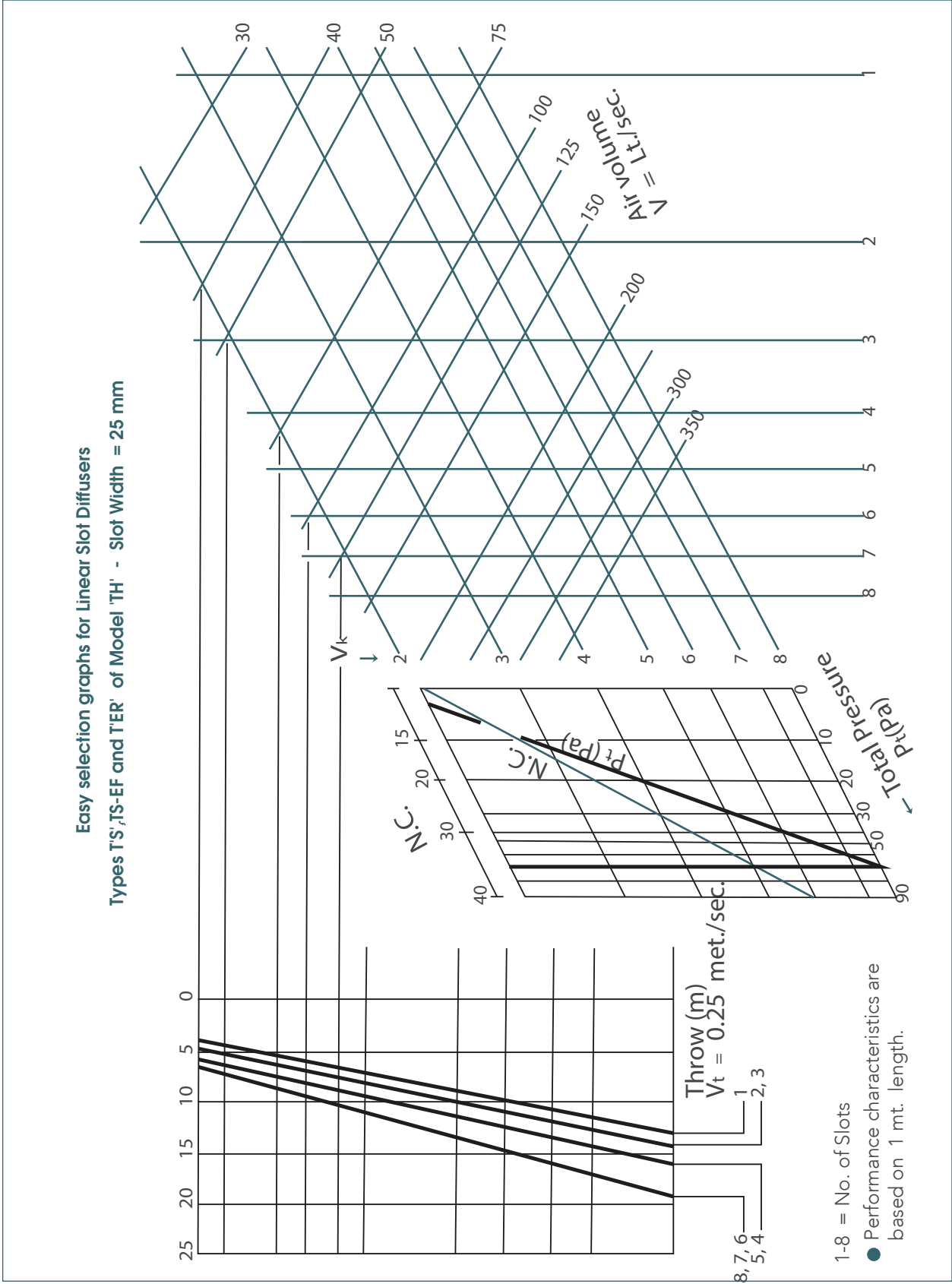
ENGINEERING AND PERFORMANCE DATA



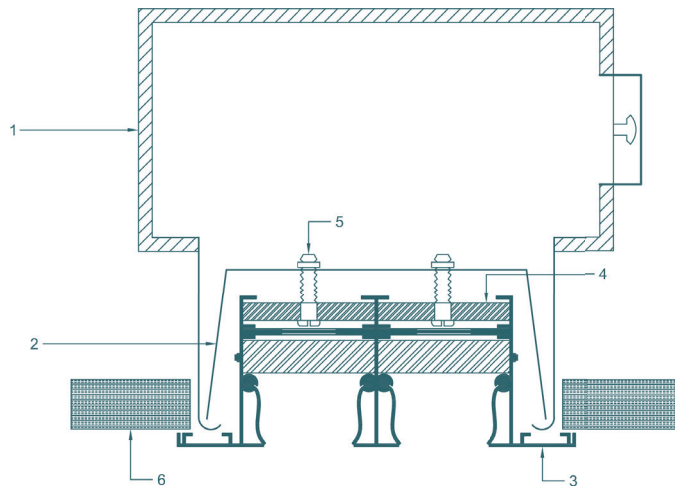
ENGINEERING AND PERFORMANCE DATA



ENGINEERING AND PERFORMANCE DATA



LINEAR SLOT DIFFUSERS FIXING DETAILS

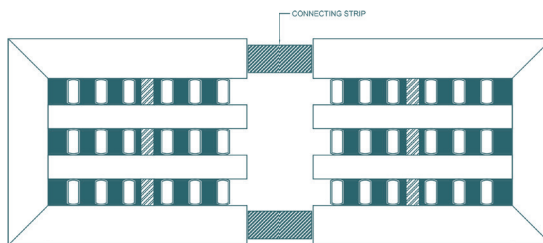


Type 'C' fixing:

Concealed bracket type fixing

1. Plenum
 2. Hanging Bridge bracket
 3. Slot diffuser
 4. Supporting clamps
 5. Adjustable screw
 6. False ceiling
- Recommended for ceiling and sill applications.

Two Long pices linear slot diffuser joining diagram



Available finishes:

- Natural anodised aluminium
- Powder coated colour finish.
(Mention RAL colour number)
- Golden anodised aluminium
- Bronze anodised aluminium
- Mill finish aluminium

Available finishes for the accessories:

Mill finish aluminium

Powder coated in black matt finish

Ordering Data:

Specify

- | | | |
|----------------------------|-------------------|-------------------|
| 1. TECNALCO type and model | 3. Quantity | 5. Surface finish |
| 2. Nominal size - L x W | 4. Type of fixing | 6. Remarks |

NOTE:

- Standard supply is Type 'C' fixing.
- Counter sunk screw type fixing is also available on request.
- Alignment or joining clips are provided for all standard type sections of TECNALCO model Linear Slot Diffusers, when supplied with open ends for continuous installation

Example:

1.	2.	3.	4.	5.	6.
SLD'TS' -TM-EA	3000 x 3 slot	25 nos.	Type 'C' fixing	Powder coated finish RAL 9016	D → Damper Black matt

NOTE:

- Supply with insulated plenum boxes of 1500 mm lengths.